right through the galvanometer.

7	**(a)**	gravitational force exerted by Sun on Earth provides the centripetal force	B1
		$m\,r\,\omega^2 = GMm / r^2$	
		$r^3 = GM \times (T / 2\pi)^2$	C1
		$r^3 = 6.67 \times 10^{-11} \times 2.0 \times 10^{30} \times (365 \times 24 \times 3600 / 2\,\pi\,)^2$	C1
		1 AU = r = 1.498×10^{11} m	
		= 1.50×10^{11} m	A1

(b)(i)	The mass of the Sun is much bigger compared to the total mass of all the planets, and so their gravitational influence is small.	B1
(b)(ii)	To escape to infinity, Initial KE of object at Jupiter ≥ gain in GPE from Jupiter to infinity	C1
	$(1/2)mv^2 \geq 0 - (-GMm / r)$	
	$(1/2)\ v^2 \geq GM / r$	C1
	$v^2 \geq (2 \times 6.67 \times 10^{-11} \times 2.0 \times 10^{30})\ / (5.2 \times 1.50 \times 10^{11})$	
	$\boldsymbol{v_{min}} = \mathbf{1.85 \times 10^4\ m\ s^{-1}}$**.**	A1
(b)(iii)	According to Fig 7.3, the speed of Voyager 2 at Jupiter on 9 July 1979 was 10.5 km s^{-1} = 1.05×10^4 m s^{-1}, which is less than the escape speed of 1.85×10^4 m s^{-1} needed at that distance from the Sun. Hence, it was not travelling fast enough to escape to infinity.	B1
(b)(iv)	From Fig. 3, the speed of Voyager 2 increased from 10.5 km s^{-1} to 28.0 km s^{-1} during its interaction with Jupiter.	
	Hence, its gain in momentum = $\Delta p = p_f - p_i = mv_f - mv_i$	C1
	$= 773\ (28.0 - 10.5) \times 10^3$	
	$= \mathbf{1.35 \times 10^7\ kg\ m\ s^{-1}}$	A1
	Velocity must be read to ½ square (1 dp)	
(b)(v)	Voyager 2 gains kinetic energy/momentum from Jupiter's orbital kinetic energy about the Sun. In a gravity assist, the spacecraft exchanges momentum/energy with the moving planet, the craft gains a tiny amount of the planet's orbital energy	C1
(c)(i)	Half-life, $t_{1/2}$ = 87.74 years	
	$A = A_0 \exp(-\lambda t)$ or $A = A_0 \exp(-\ln 2 \times t / t_{1/2})$	C1
	$A = A_0 \exp(-\ln 2 \times 1/87.74)$	
	$A = A_0 \exp(-0.00790)$	C1
	Activity remaining after one year is $A = 0.992\ A_0$	
	Hence, fractional decrease in activity in one year = 1 – 0.992 = 0.0079 = **0.79 %**	A1
(c)(ii)	$A = 0.623\ A_0$	
	Power output is proportional to the activity, so	B1
	$P = 0.623\ P_0$	
	$P = 0.623 \times 470 =$ **290 W**	A1
(c)(iii)	As spacecraft moved farther away, the intensity of the received radio waves had decreased, power received decreases (since power received = Intensity at that point x area of receiver.)	B1
	Hence, the total area of the receiving parabolic dish antenna needed to be increased so that the total power of the received waves will be large enough to be detected.	B1

(d) In 1990, distance of Earth from Voyager 1 = 40.47 AU

Angular diameter of Earth = $\theta = d$ / D C1

$= (2 \times 6.4 \times 10^6) / (40.47 \times 1.50 \times 10^{11}) = \mathbf{2.1 \times 10^{-6}}$ **rad** A1